

Introduction

In this practical session, you will continue the work from TP01. In lesson 1, you practiced how to organize information into clear and efficient structures. In lesson 2, you will now represent data using relations and write simple relational algebra expressions to query data.

Description

For each exercise below, read the scenario and do the requested work. Depending on the exercise, you may need to:

- write one or more relation schemas
- identify attributes in each relation
- represent links between relations clearly
- write relational algebra expressions using standard operators

You can answer using relation schema format, small tables, and relational algebra notation such as σ , π , $\bowtie$, $\cup$, $\cap$, and $-$.

Learning Objectives

- Convert structured information into relations
- Identify attributes and common attributes between relations
- Understand how relations are connected
- Apply basic relational algebra operations

Submission Guidelines

- Write one answer for each exercise
- Use consistent relation and attribute names
- Identify attributes clearly when needed
- Write relational algebra expressions carefully and correctly

1. Exercise 1 - Fruit Relation

In TP01, you organized information about fruits in a grocery shop. The shop has apples, bananas, and oranges. There are ten apples, and three of them are rotten while the rest are fresh. There are five bananas, and all of them are fresh. There are eight oranges, of which two are rotten and the rest are fresh.

Your task is to convert this grouped information into a relation that stores the data clearly with no unnecessary repetition.

2. Exercise 2 - Students and Courses

A university stores information about students and courses. Each student has a student ID, name, city, and age. Each course has a course ID, course name, and credit value.

Your task is to design the relations for these two entities and list the attributes of each relation.

3. Exercise 3 - Departments and Teachers

A university manages departments and teachers. Each department has a department ID, department name, and office. Each teacher has a teacher ID, teacher name, degree, and department ID. Each teacher works in one department, and one department can have many teachers.

Your task is to design the relations for this situation and identify the common attribute that connects the two relations.

4. Exercise 4 - Student Enrollment

Students can enroll in multiple courses, and each course can also contain many students. Assume you already have the relations `Student(StudentID, Name)` and `Course(CourseID, CourseName)`.

Your task is to design the additional relation needed to represent enrollment and show how it links students and courses.

5. Exercise 5 - Relational Algebra

Consider the following relations:

`Student(StudentID, Name, City)`

StudentID	Name	City
S01	Dara	Phnom Penh
S02	Sophea	Siem Reap
S03	Vanna	Phnom Penh

`Course(CourseID, CourseName)`

CourseID	CourseName
C1	Math
C2	Physics

`Enrollment(StudentID, CourseID)`

StudentID	CourseID
S01	C1
S01	C2
S02	C2

Write relational algebra expressions for the following questions:

- Find students who live in Phnom Penh
- Show only the names of students
- Show each student with the course IDs they take
- Show each student name with the course names they take
- Find students who take Physics